LANGUAGE PATTERNS IN AI VS HUMAN SCIENTIFIC ABSTRACTS

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ABSTRACT

The growth of Large Language Models (LLMs) in academia has created a need for clear methods to tell human-authored scientific text from machine-generated writing. While complex neural classifiers achieve high accuracy, the tools do not explain decisions and can be bypassed by minor edits. This study examines whether a small set of easy-to-understand features from language study and style analysis can identify the writing style of machine-generated scientific abstracts. A balanced dataset of 200,000 academic abstracts was analyzed, created by collecting 100,000 human-written abstracts from the OpenAlex academic database (all pre-2022 to ensure a clean human baseline), then generating a matching AI abstract for each human abstract using GPT-4o-mini, yielding 200,000 total abstracts after cleaning. Eight features were measured, including vocabulary variety, sentence length variation, and the use of connecting or signaling words. The analysis indicates that AI-generated text has less variety in sentence length, uses more first-person pronouns like "we", and shows higher lexical diversity than human writing. A classifier using these features achieved an accuracy of 95.69% with an AUC-ROC of 0.991, with lexical diversity as the most important feature. These findings suggest that AI-generated scientific text has a distinct writing style that can be detected using interpretable language features.

1 INTRODUCTION

The release of consumer-accessible Large Language Models has changed academic publishing. While these tools offer helpful support for language editing and brainstorming, the ability to generate clear and well-structured scientific text has raised concerns about academic honesty, fake paper generation, and the overall quality of research papers [4; 5]. In the past, writing a scientific paper required deep knowledge of the subject and years of practice in academic writing. Now, anyone with access to an advanced computer program can produce text that looks like a professional scientific document. This change forces the academic community to rethink how peer review works and how originality is checked. The ease of generating text means that conferences and journals receive more submissions than ever before, but reviewers struggle to tell which papers are authentic.

Current detection tools fall into two main groups, which are deep learning classifiers and probability indicators. Although these tools perform well on standard test sets, the tools have two main drawbacks.

1. **Lack of Interpretability.** These tools do not explain decisions. This makes the tools difficult to use when an institution needs to make a formal decision about a student or researcher [6]. A tool that cannot explain why a text looks generated has limited value in hearings. When a student is accused of academic dishonesty, a simple % score from a machine learning model is not enough evidence. Committees need clear examples of how the writing differs from normal human patterns. Without explainable evidence, universities hesitate to enforce rules because of fears about false accusations and legal challenges. This creates a gap between the availability of detection tools and actual usefulness in real-world settings.
2. **Adversarial Vulnerability.** Simple edits, paraphrasing, or clever prompts can easily bypass these probability detectors [7]. As language models improve, detection tools that depend

on statistical patterns face constant challenges. A student can simply ask the model to rewrite the text to avoid detection, and the software may fail to flag it. Different language models also leave different statistical traces. A detector trained on text from one specific model might completely fail to recognize text from a newer system. This creates a constant race where detection software is always one step behind the latest generation tools. The academic community needs a more stable method of identifying generated text that does not rely on hidden mathematical patterns.

This study uses an alternative approach based on style analysis and language study. Instead of using complex neural classifiers, this work examines interpretable features such as vocabulary variety, sentence rhythm, and the use of signaling words to map the style of machine-written scientific text. Drawing on Hyland's framework [2] and style analysis methods [1], this study examines a balanced dataset of 200,000 abstracts to find the differences between human and machine writing, offering a clear and explainable model for verifying scientific texts.

Human writing is shaped by cognitive constraints. When people write, ideas are held in working memory and translated into words one sentence at a time. This mental effort creates a natural variation in sentence length and structure. Some sentences are short and direct to convey key points, while others stretch out to connect complex ideas. Machines, however, produce text by calculating the most likely next word based on patterns learned from large text corpora [15]. Machines do not experience cognitive load or fatigue. As a result, machine writing often takes on a uniform rhythm that human readers can sense but struggle to define. By measuring these specific style choices, this research provides a concrete way to describe that artificial rhythm.

The goal of this paper is not to build a perfect detection system that catches every generated sentence. Instead, the focus is on understanding the basic differences in how humans and machines approach the task of scientific writing. Identifying these specific linguistic markers provides educators and reviewers with practical tools to evaluate suspicious texts. When a reviewer understands what makes a text look artificial, the reviewer can rely on personal judgment rather than trusting complex software programs. This study provides the data needed to build that understanding, based on a large sample of modern academic abstracts.

2 Literature Review

AI-Generated Text Detection in Science. The challenge of identifying machine-generated text in academic settings is not entirely new. Early research on machine-generated scientific writing focused on basic tools like SCIgen, which produced grammatically correct but meaningless text by randomly selecting words from a predefined list [8]. These early generators were easy for human readers to spot because the sentences lacked logical flow and the ideas did not make sense together. Modern language models, however, write text that is clear, coherent, and fits the academic style perfectly. Research shows that human peer reviewers struggle to identify AI-generated abstracts, getting results close to random chance when trying to tell them apart from human writing [4]. This difficulty has led to the rapid development of automatic detection systems designed to help reviewers and publishers maintain the quality of scientific literature.

The most common approach to solving this problem uses pre-trained language models as classifiers. Guo et al. (2023) showed that these deep learning classifiers can achieve over 95% accuracy on similar data but perform much worse when tested on text from new models or different academic topics [5]. Another approach relies on probability. Mitchell et al. (2023) introduced a zero-shot method, meaning the detector works without being trained on labeled examples first. Instead, the method analyzes probability patterns in the text directly, but the method requires access to the model's internal settings, which are often kept secret by the companies that build the models [9]. More recently, Hans et al. (2024) introduced a reference-free method that compares patterns from two different language models to detect AI writing [10]. While these systems are technically impressive, the systems share a common flaw. As text generation improves, the statistical signals the systems rely on become harder to find. This makes it useful to build detection methods based on established language theory, where the measured features represent basic choices in writing rather than temporary model settings.

Stylometrics and Lexical Diversity. Style analysis, or stylometrics, is the statistical study of linguistic style. This field assumes that every writer has a unique, subconscious writing pattern that remains somewhat consistent across different texts [1; 14]. Historically, researchers used stylometrics to figure out who wrote anonymous historical documents or to settle disputes over literary authorship. Key features include sentence length patterns, the frequency of common function words, and overall vocabulary wealth. Biber (1988) showed that different writing styles are best captured by groups of features working together rather than by looking at single words in isolation [1]. This multi-feature approach has become the standard for analyzing complex texts.

In human writing, vocabulary variety is naturally limited by the writer's personal vocabulary and the specific needs of the topic being discussed [3]. Measuring this variety, however, is a well-known mathematical challenge. Common metrics for vocabulary richness are often negatively affected by text length, giving unfairly lower scores to longer texts simply because words inevitably repeat as the document grows. The Measure of Textual Lexical Diversity (MTLD) solves this exact issue by computing scores over sequential text segments rather than the whole document at once. This provides a reliable measure of vocabulary variety for short texts like abstracts [11; 18]. Recent studies have started to look at vocabulary variety in machine text. For example, Guo et al. (2023) found that AI-generated abstracts sometimes use a more uniform set of words than human writing, depending heavily on the specific prompt and the model settings used during generation [5].

Metadiscourse Frameworks. Metadiscourse refers to the specific language choices writers make to organize text, connect with readers, and show personal attitude toward the content. Under Hyland's established model, these linguistic choices are divided into two main categories [2].

- **Interactive Resources.** These help guide the reader smoothly through the text. Interactive resources include transition words like "however," "therefore," and "furthermore," as well as examples like "for instance" and "namely." These words act as signposts that help the reader understand the logical structure of the argument.
- **Interactional Resources.** These show the writer's perspective and personal involvement. Interactional resources include hedging words like "may," "suggests," and "possibly," which authors use to show caution. The resources also include boosters like "clearly," "demonstrates," and "proves," which authors use when feeling very confident. Finally, interactional resources include self-mentions like "we," "our," and "my."

In academic writing, these choices play a central role because they define the scientific tone. Hedging words let authors express appropriate scientific caution when discussing uncertainty or early results. Boosters show confidence when the experimental evidence is strong. Self-mentions help the author establish a role in the research process and take credit for the work [2]. Comparing how machines and humans use these specific groups of words shows whether the model captures the nuances of scientific style or simply reproduces the factual content and structure.

Syntactic Rhythm and Sentence-Level Variation. The rhythm of writing, measured through sentence length variation, is another useful feature for style analysis. Skilled human writers naturally vary sentence lengths to manage information density and keep the reader engaged. Short, punchy sentences draw attention to key points and longer, more complex sentences provide detailed explanations or background context [12]. This creates a natural, breathing variety in sentence length throughout a paragraph. If language models, driven by mathematical predictions, produce sentences of very similar lengths, this lack of variety creates a monotonous rhythm that might help identify machine-written text even when the vocabulary is perfect.

3 Methodology

Corpus Construction. The foundation of this study is a balanced dataset of 200,000 scientific abstracts built to represent both authentic human writing and modern machine generation. Building a high-quality corpus is a foundational step in stylometric analysis, because any bias in the data collection process may skew the final results.

First, 100,000 human abstracts were sourced from the OpenAlex academic database [13], a free and open catalog of scholarly works. Abstracts were retrieved using the OpenAlex API with random sampling, filtering for English-language journal articles with available abstracts and publication dates before 2022. This ensures a clean human baseline unaffected by post-ChatGPT AI assistance.

This collection represents authentic, peer-reviewed scientific writing drawn from real academic journals across all scientific disciplines. To ensure data quality, the dataset went through a standard filtering process. All duplicate entries were removed to prevent the classification model from learning specific repeating phrases. In addition, very short records under one hundred words were excluded, because stylometric tools need a minimum amount of text to accurately measure features like vocabulary variety and sentence length rhythm [18].

Next, a custom generation pipeline was built to create matching AI abstracts for each of the 100,000 human paper titles using GPT-4o-mini, a widely used modern language model from OpenAI. This paired approach ensures that both the human and AI groups cover the exact same scientific topics, which prevents the classifier from simply learning topic-specific vocabulary instead of actual writing style. The generation process used parallel workers with automatic retry logic to handle rate limits and network timeouts. Each request sent a simple, neutral prompt asking the model to write an academic abstract for the provided paper title. The model was accessed via the OpenAI API in October 2024, at a temperature of 0.7 with a maximum token limit of 400. No special instructions about tone or style were given, to capture the default writing behavior of the model. The pipeline successfully generated abstracts for all 100,000 titles. After cleaning and deduplication, 100,000 paired abstracts remained. All 100,000 human and 100,000 AI abstracts were used for analysis, yielding a final dataset of 200,000 abstracts.

Feature Extraction. Once the corpus was built, the text needed to be converted into numbers that a machine learning model could understand. The linguistic features were extracted using a custom Python pipeline built around the `lexicalrichness` package (version 0.4.1) for vocabulary mathematics and regex-based tokenization for sentence and word segmentation. Eight specific features were calculated for every abstract in the dataset. The hedging, boosting, and connector word lists used in the feature extraction pipeline were compiled from established linguistic resources, primarily Hyland's published word lists [2], and applied via case-insensitive string matching. The full lists are available in the project repository.

Syntactic Features. These features measure the basic physical structure of the text.

1. Mean Sentence Length. The average number of words per sentence. This measures how dense the information is packed.

2. Sentence Length Coefficient of Variation (CV). This measures the mathematical variation in sentence lengths across the text. Higher values mean the writer uses a mix of very short and very long sentences, creating a varied and natural rhythm. Lower values mean all sentences are roughly the same length.

Lexical Features. These features measure vocabulary usage.

3. Lexical Diversity (MTLD). A mathematical measure of vocabulary variety that is uniquely designed to not be affected by text length [11; 18]. While simpler metrics like Type-Token Ratio (TTR), which simply divides the number of unique words by the total word count, and Yule's K, which estimates vocabulary richness based on word frequency distributions, are heavily biased by the total length of the text, the Measure of Textual Lexical Diversity (MTLD) solves this issue by computing average segment lengths over varying diversity thresholds. This provides a reliable, unbiased measure of vocabulary variety for short texts like abstracts.

Metadiscourse Features. These features measure how the writer guides the reader, calculated as counts per 1,000 words to ensure fair comparison between texts of different lengths.

4. Hedge Density. The frequency of cautious words like *may, might, possibly, suggests*.

5. Booster Density. The frequency of confident words like *clearly, demonstrates, establishes*.

6. Self-Mention Density. The frequency of first-person pronouns like *we, our, my*.

7. Connector Density. The frequency of logical transition words like *however, moreover, therefore*.

8. Sentence-Opener Connector Ratio. The proportion of sentences that explicitly start with a transition word, which measures how heavily the writer relies on obvious structural signposts.

Statistical Testing. To find meaningful differences between the human and AI groups before building a classifier, this study used Mann-Whitney U tests. These tests are highly suitable for linguistic data because text features rarely follow a perfect normal bell curve, violating the core assumptions of parametric tests like Student's t-test. The Mann-Whitney U test checks whether one group tends to have higher values than the other without assuming any specific distribution shape. However, a statistically significant p-value only tells us that a difference exists, not how large or meaningful that difference is. With 200,000 texts, even tiny differences can appear statistically significant. To address this, Cohen's d was calculated. Cohen's d is a measure of effect size that expresses the difference between two groups in terms of standard deviations. A value of d = 0.2 means the groups overlap by about 85% and is considered a small effect. A value of d = 0.5 means the groups overlap by about 67% and is considered a medium effect. A value of d = 0.8 means the groups overlap by about 53% and is considered a large effect. Values above d = 1.0 mean the groups overlap by less than 45%, indicating a very clear and obvious difference. In this study, a large effect size means the feature can reliably separate human and AI texts in practice, not just in theory.

Classification Model. To test the combined predictive power of these eight features, a Random Forest classifier was trained using scikit-learn (version 1.6.1) [16]. Random Forest was chosen because it handles non-linear relationships well and is highly resistant to overfitting, which is when a model memorizes the training data instead of learning general patterns, unlike complex neural networks. The model used 100 decision trees (n_estimators=100) with all other parameters at their default values. The model was trained and tested using a strict 5-fold stratified cross-validation setup with a fixed random seed of 42 for reproducibility. This means the data was split into five equal parts, with each part containing the same proportion of human and AI texts, and the model was trained five separate times to ensure the results were stable and reliable. The performance was evaluated using multiple metrics. Accuracy measures the overall % of correct predictions. Precision measures what % of texts flagged as AI were actually AI. Recall measures what % of actual AI texts were successfully caught. F1 score is the balanced average of precision and recall. AUC-ROC (Area Under the Receiver Operating Characteristic curve) measures how well the model separates the two groups, where a value of 0.5 means random guessing and 1.0 means perfect separation. The model also measured feature importance, showing how much each feature contributed to the final classification decisions.

4 RESULTS

Distributional Analysis. Table 1 compares human and AI abstracts across the eight extracted features. All eight features show statistically meaningful differences at the $p < 0.001$ level, indicating that the two groups of texts have distinct mathematical profiles.

Feature	Human Mean (SD)	AI Mean (SD)	p-value	Cohen's d
Lexical Diversity (MTLD)	75.1128 (31.8084)	118.6241 (28.0710)	< 0.001	1.4505
Sentence Length CV	0.4604 (0.2737)	0.2324 (0.1156)	< 0.001	-1.0852
Self-Mention Density	6.8968 (11.2442)	16.2204 (10.5869)	< 0.001	0.8538
Sentence-Opener Connector Ratio	0.0280 (0.0670)	0.0967 (0.0646)	< 0.001	1.0435
Connector Density	3.5322 (5.3211)	6.1125 (3.7631)	< 0.001	0.5599
Hedge Density	3.3302 (6.0175)	5.4220 (4.9898)	< 0.001	0.3784
Mean Sentence Length	21.4772 (16.7702)	23.8430 (3.4171)	< 0.001	0.1955
Booster Density	3.1246 (5.4744)	3.4405 (3.5770)	< 0.001	0.0683

Table 1: Statistical Comparisons between Human and AI Abstracts (N = 200,000)

The largest difference appears in lexical diversity (MTLD), where AI abstracts show a much higher mean (118.62) than human abstracts (75.11), with a very large effect size (d = 1.45). This finding differs from earlier studies of older models like ChatGPT (GPT-3.5), which found AI text to be lexically more uniform [5]. The difference may reflect improvements in newer models like GPT-4o-mini, which appear to draw from a broader vocabulary pool. Sentence length variation (CV) shows the second largest difference. Human abstracts have a much higher mean variation (0.4604) than AI abstracts (0.2324), with a very large effect size (d = -1.09). The standard deviation for human abstracts (0.2737) is more than double that of AI abstracts (0.1156). This wider spread in human writing suggests that human authors have a diverse range of personal writing styles, whereas language models tend to cluster tightly around a single, default rhythm. The human distribution also shows outliers with very high self-mention density values, which likely reflect specific disciplinary conventions where first-person pronouns are standard, such as computer science and mathematics.

These outliers do not appear in the AI distribution, suggesting that the models produce a more uniform pattern regardless of disciplinary context. Figure 1 illustrates these distributions using overlaid density plots, which show how frequently each value occurs for both groups. The width of each curve at any point indicates how many texts have that value, making it easy to see where the data clusters and where it spreads out. The AI density curve for sentence length CV is narrow and peaked, while the human curve is wide and spread out across the entire range.

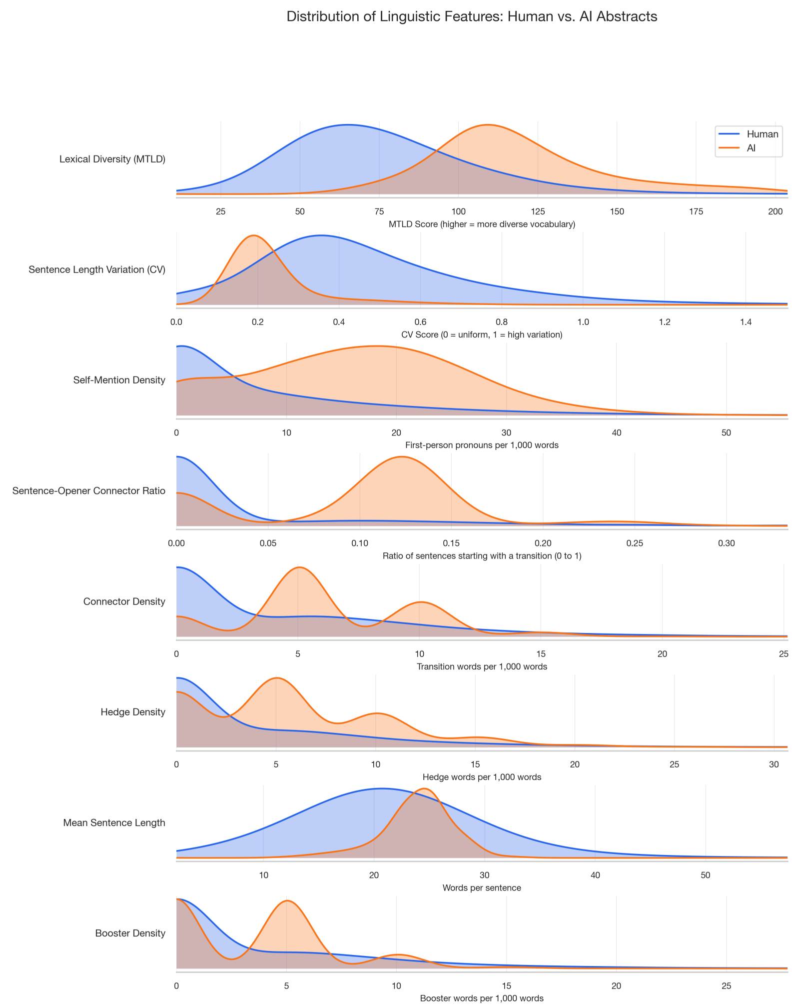

Figure 1: Distribution of linguistic features across human and AI abstracts. Each row shows overlaid kernel density estimates for human (blue) and AI (orange) texts. The sentence length CV row shows the clearest separation, with human abstracts exhibiting broader distributional spread and higher central tendency.

Figure 2 visually breaks down the effect sizes for all eight linguistic features using Cohen's d metric. This chart is useful because a p-value only indicates if a difference exists, but Cohen's d indicates how practically important that difference actually is. As the chart shows, sentence length variation has a very large negative effect size, while lexical diversity and self-mention density show very large positive effect sizes, meaning AI text tends to use more diverse vocabulary and more first-person pronouns compared to humans in this dataset. The sentence-opener connector ratio also shows a very large effect ($d = 1.04$), indicating that AI text relies heavily on starting sentences with transition words. The remaining features, such as connector density and hedge density, show medium and small but reliable differences.

Classification Performance. Building on these individual differences, the next step was to see if a machine learning model could combine these features to reliably separate human text from AI text. Table 2 presents the detailed performance metrics of the Random Forest model across the 5-fold

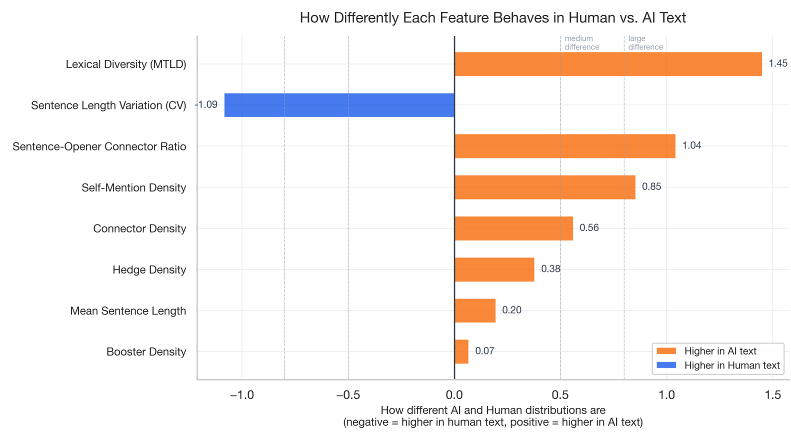

Figure 2: Cohen's d effect sizes for all eight linguistic features. Blue bars indicate features with higher values in human text; orange bars indicate features with higher values in AI text. Dashed lines mark conventional thresholds for medium and large effects.

cross-validation tests.

Metric	Value
Accuracy	0.9569 ± 0.0012
Precision	0.9618 ± 0.0014
Recall	0.9515 ± 0.0026
F1 Score	0.9566 ± 0.0013
AUC-ROC	0.9906 ± 0.0003

Table 2: Classification Performance Metrics (5-Fold Stratified Cross-Validation)

The classifier achieves a stable overall accuracy of 95.69% under the controlled conditions of this study. The precision and recall scores are well balanced at approximately 95 to 96% each. This balance suggests that the model is roughly equally effective at identifying AI text and recognizing human text, rather than favoring one category over the other. However, this balance was achieved on a dataset where AI texts were generated with a single, simple prompt and not subsequently edited. The extremely low standard deviations across all metrics (all below 0.003) further suggest that these results are reproducible, though generalization to other models, prompts, or edited texts remains untested.

Feature Importance. Knowing that the model works, the most important question is how it makes its decisions. Table 3 breaks down the relative importance of each feature in the Random Forest classification model.

Feature	Gini Importance
Lexical Diversity (MTLD)	0.2317
Sentence Length CV	0.2002
Connector Density	0.1424
Sentence-Opener Connector Ratio	0.1323
Self-Mention Density	0.0985
Hedge Density	0.0786
Mean Sentence Length	0.0688
Booster Density	0.0475

Table 3: Random Forest Feature Importance

Lexical diversity is the most important feature, accounting for 23.17% of the model's decision-making power. This supports the earlier statistical tests showing the largest effect size (d = 1.45) for this feature. Sentence length variation follows at 20.02%, suggesting that the physical rhythm of the text may be a useful signal for identifying machine writing. Figure 3 visually ranks these features. Connector density (14.24%), sentence-opener connector ratio (13.23%), and self-mention density (9.85%) form the second tier of important features. The metadiscourse features like hedging and boosting, while statistically different in Table 1, only play a minor supporting role in the actual classification model, each contributing less than 8% to the final decisions.

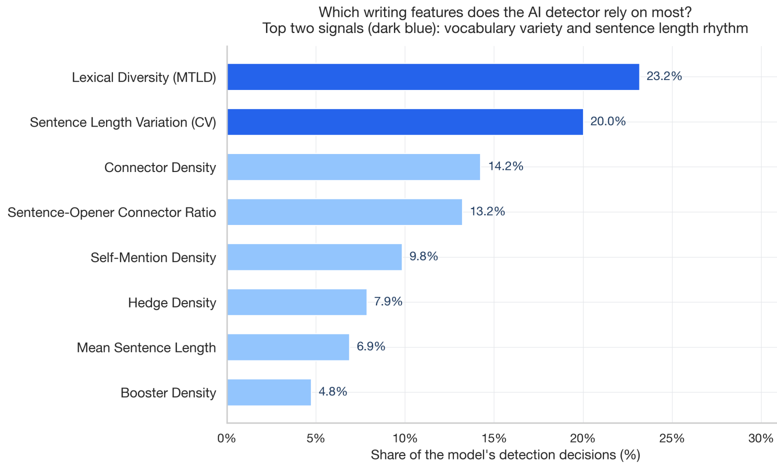

Figure 3: How much each writing feature contributes to the model's detection decisions. Darker bars are the two primary signals the detector relies on most; lighter bars are supporting signals. Each bar shows the percentage share of the model's total decisions.

Figure 4 summarises the model's performance across five key measures. The overall accuracy of 95.7% means roughly 1 in 23 abstracts is misclassified. Precision and recall are closely matched, indicating the model makes roughly equal numbers of false alarms and missed detections rather than being biased toward one error type. Figure 5 shows the raw counts from a separate held-out test set.

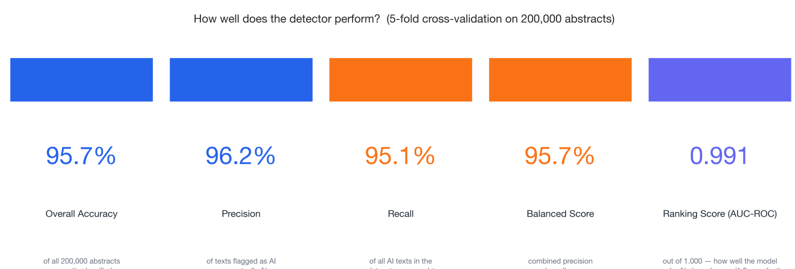

Figure 4: Model performance scorecard from 5-fold cross-validation on 200,000 abstracts. Each card shows one measure: overall accuracy, how often AI flags are correct (precision), how many AI texts are caught (recall), the combined balanced score, and the ranking score (AUC-ROC) out of 1.000.

Feature Independence. To ensure the model was not just measuring the exact same thing eight different times, Figure 6 maps the correlation among all features. A good classification model needs independent features that each bring unique information. As the heatmap shows, most features have very weak correlations (scores close to zero). This suggests that the features likely measure different aspects of writing style. The only strong correlation (0.65) is between overall connector density and the ratio of sentence-opening connectors, which makes logical sense since the two are mathematically related formulas. This independence suggests that the eight chosen features may capture a broad, well-rounded picture of an author's style, at least within the constraints of this dataset.

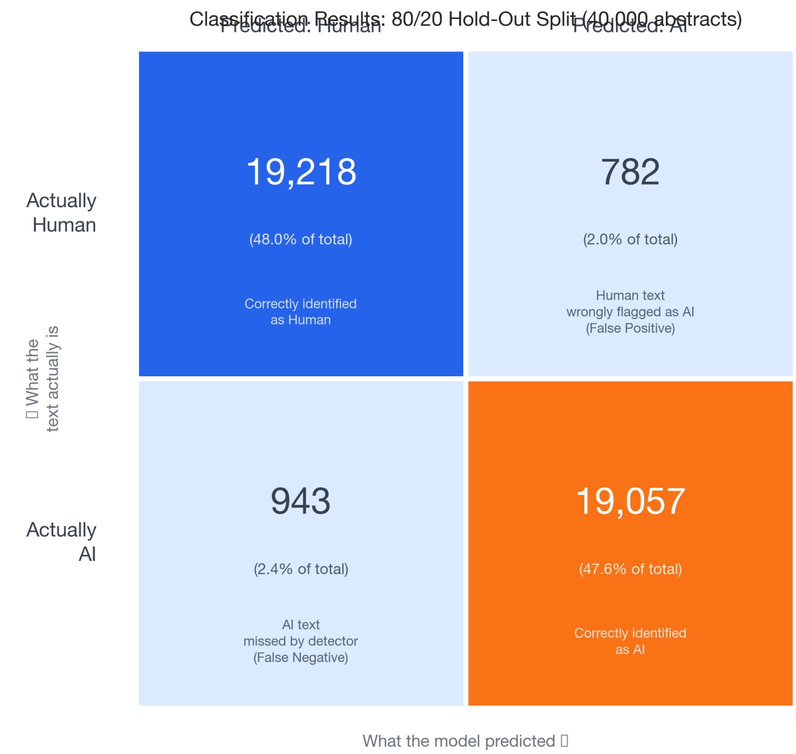

Figure 5: Classification results on 40,000 held-out abstracts (80/20 split). Dark cells on the diagonal show correct predictions; light cells show errors. The model correctly identified 96.1% of human abstracts and 95.3% of AI abstracts.

5 Discussion

Sentence Length Variation as a Primary Marker. One of the key findings of this study is the highly uniform sentence length found in AI-generated scientific abstracts. This structural uniformity may serve as a useful, mathematically measurable indicator of machine writing. Human writers, even in highly constrained academic formats, tend to vary sentence structures. Short, declarative sentences are often used for emphasis, summarizing key findings, or making bold claims. Longer, clause-heavy sentences are typically used to explain complex methodologies, introduce nuanced background context, or outline detailed limitations [12]. This natural variation creates a varied rhythm in human writing that may help keep the reader engaged and manage the flow of dense scientific information.

AI generators, however, tend to produce sentences of unusually similar lengths, resulting in a flatter, more uniform pacing that may serve as an indicator of machine authorship. This difference likely stems from the basic architecture of how human and machine language production works. Human writing is an intentional, structured process shaped by working memory limits and deliberate planning [12]. A human author thinks about the overall paragraph structure and adjusts sentence length to fit the logical flow of ideas. Large language models, on the other hand, generate text autoregressively, meaning the model produces one word at a time by predicting the most likely next word based on all the words that came before it [15]. Each word is a token, which is the smallest unit of text the model processes. Because the models optimize for mathematical probability rather than intentional rhetorical structure, the output tends to regress to the mean, meaning sentence lengths may cluster around a safe, average length rather than varying widely. The models may not perform the kind of high-level structural planning needed to purposely insert a five-word sentence followed by a forty-word sentence for rhetorical effect, though this may change as models evolve.

Self-Mention Rates and Academic Style. An unexpected finding was that AI abstracts in this dataset contain more than double the rate of first-person pronouns (such as “we,” “our,” and “my”) compared to genuine human abstracts. This suggests that modern language models may frequently

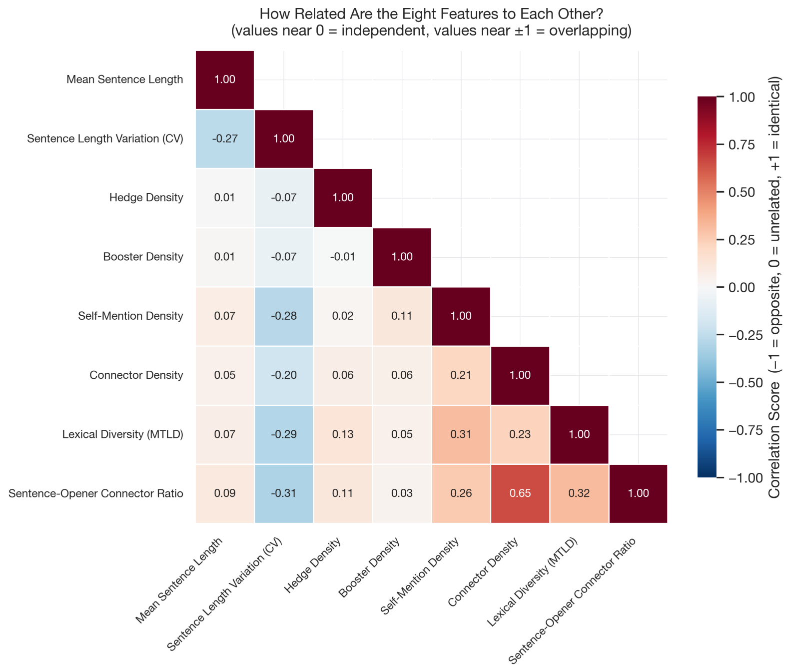

Figure 6: How correlated the eight features are with each other (lower triangle). Most pairs score near 0, meaning they measure independent aspects of writing style. The one exception is connector density and sentence-opener connector ratio (score = 0.65), which overlap because they are mathematically related.

use self-mentions to simulate an academic style, potentially overusing this feature compared to actual human researchers.

In real academic journals, the use of first-person pronouns is strictly governed by the conventions of the specific scientific field [2]. Some disciplines, like computer science and mathematics, frequently use "we" to guide the reader through a proof or algorithm ("we assume," "we define"). However, many other fields, particularly in the hard physical sciences and medicine, traditionally discourage first-person pronouns in favor of the passive voice to maintain an objective, impersonal tone ("it was observed," "the samples were tested"). The notably high rate of these words in machine-generated text may occur because language models are trained on massive, mixed corpora that blend materials from all scientific disciplines. When prompted to write an "academic abstract," the model may pull stylistic features from this blended data without accounting for the subtle, field-specific rules governing when self-mention is actually appropriate.

Hedging, Boosting, and Scientific Stance. The analysis of metadiscourse features revealed that AI abstracts use more booster words and more hedging words than authentic human writing. While AI text uses more hedges in absolute terms, the difference is smaller than expected, suggesting that GPT-4o-mini has learned to mimic some degree of scientific caution. However, the overall pattern suggests that AI text tends to be more assertive and confident, particularly in the use of boosters and connectors.

This pattern may be tied to the commercial optimization of these models. Language models are typically fine-tuned by their creators to be helpful, confident, and fluent using reinforcement learning from human feedback [17], which may inadvertently lead to highly assertive statements. The model tends to provide a strong, clear answer. Professional human researchers, however, are trained to be rigorously cautious. Human researchers understand that a single study rarely "proves" anything absolutely, and therefore tend to rely heavily on hedges like "suggests," "indicates," or "may contribute to." This cautious nuance appears to be a hallmark of professional scientific work that current language models may struggle to accurately replicate without very specific prompting [5].

Lexical Diversity Findings. In contrast to earlier findings with older models [5], this large-scale analysis found a very large and meaningful difference in lexical diversity (MTLD) between human and AI abstracts. AI-generated abstracts show substantially higher lexical diversity (mean = 118.62) compared to human abstracts (mean = 75.11), with a very large effect size (d = 1.45). This finding, based on 200,000 texts, suggests that modern language models like GPT-4o-mini may draw from a broader vocabulary pool than typical human writers when composing scientific abstracts.

This result likely reflects the training approach of modern language models. GPT-4o-mini has been trained on vast corpora spanning diverse scientific disciplines, which may give the model access to a wider active vocabulary than most individual human researchers, who tend to specialize in a narrow subfield. Human writers naturally fall into repetitive patterns within domain-specific terminology, while the model tends to vary word choices more broadly, producing text that sounds fluent and natural. The Random Forest model assigned the highest importance weight (23.17%) to this feature, suggesting that lexical diversity may be the most powerful diagnostic signal when measured at scale, at least for this model and prompt configuration.

Practical Application for Academic Integrity. These results may offer practical value for publishers, universities, and academic integrity systems, though with important caveats. The potential value lies in the nature of the features themselves. Unlike current "black-box" neural network detectors [5; 6], which provide only a probability score, the eight features identified in this study are human-readable and individually interpretable. This distinction matters because academic misconduct hearings require evidence that can be examined, questioned, and defended [6]. When a tool flags a text as AI-generated but cannot explain why, the institution faces a difficult position. The tool may be correct, but the institution cannot demonstrate the basis for its decision to a student or review board. The stylometric features identified here, by contrast, could potentially allow an institution to point to specific, measurable writing characteristics, such as an unusually low sentence length variation or an abnormally high self-mention density, and explain why these patterns are inconsistent with typical human writing in that field.

However, several limitations temper this practical application. The 95.69% accuracy achieved in this study, while promising, means that approximately 4 out of every 100 texts may be misclassified. In an academic integrity context, even a small false positive rate could have serious consequences for students or researchers who are wrongly accused. Furthermore, the features identified here are based on a single model (GPT-4o-mini) and a single prompt type, and it remains uncertain whether these same patterns would hold across different language models, different prompting strategies, or texts that have been edited by humans after generation. Therefore, these features should be viewed as potential supplementary evidence rather than standalone proof of AI authorship.

The near-perfect AUC-ROC of 0.991 suggests that interpretable linguistic features, when measured on a large and controlled corpus, can potentially match the discriminative power of black-box approaches [6]. However, this performance was achieved under controlled conditions where AI texts were generated with a single prompt and not subsequently edited. Real-world scenarios, where users may paraphrase, edit, or intentionally vary their writing style, would likely reduce this performance. The purpose of this work is therefore not to replace neural detectors, but rather to suggest that explainable, transparent linguistic features may serve as a useful complement to existing tools, particularly in contexts where understanding the basis for a detection decision is as important as the decision itself.

Limitations and Future Work. While these findings are stable within the conditions tested, this study has several limitations that must be acknowledged. First, the AI abstracts were generated using a single, neutral prompt with no subsequent editing. This means the features identified here capture the default writing behavior of the model, not the full range of what AI can produce. When a language model is given different instructions, such as being told to write with varied sentence lengths, to use hedging language, or to avoid first-person pronouns, the resulting text may not show the same stylistic patterns. Similarly, when AI-generated text is revised by a human who changes the tone, style, or structure, the final output may shift away from the default AI pattern and look more like human writing. In such cases, the stylometric features identified in this study may fail to detect the text as AI-generated. This adversarial vulnerability means that stylometric detection tools must be continuously updated as language models evolve and as users learn to manipulate outputs [7].

Second, the dataset includes papers from dozens of different scientific fields but does not statistically control for differences between fields. Writing styles, particularly the use of metadiscourse and self-mention, vary radically by discipline [2]. A future study could analyze these features within specific fields (e.g., comparing only biology papers to biology papers) to create more precise baselines. Third, the human abstracts are restricted to pre-2022 publications to ensure a clean human baseline unaffected by post-ChatGPT AI assistance. Fourth, the model used here (GPT-4o-mini) represents a specific generation of AI technology. As language models continue to evolve, the models may eventually be trained specifically to mimic human sentence length variation or field-specific hedging. Continuous monitoring and updating of these stylistic baselines may be necessary.

Fifth, the human baseline corpus draws from OpenAlex without controlling for author language background. Non-native English writers may produce stylometric profiles that differ from native speakers within the same field, which could affect both the baseline measurements and the classifier's error rate for that subpopulation. Future work should examine whether these features behave differently across author language backgrounds.

Sixth, this study does not test the detection system on mixed-authorship texts where human writers have edited AI-generated drafts or vice versa. The stylistic signals identified here may be weaker in such hybrid scenarios, which are increasingly common in real-world academic writing. Future work should test whether these features remain reliable when AI text is generated with diverse prompts, multiple rounds of revision, or explicit instructions to mimic human writing patterns.

6 Conclusion

This study analyzed a corpus of 200,000 scientific abstracts to identify the core stylometric writing features that distinguish AI-generated abstracts from human-written abstracts. The statistical results indicate that, within this dataset, text generated by modern language models tends to be characterized by highly uniform sentence lengths, higher rates of first-person pronouns, elevated lexical diversity, and a greater reliance on sentence-opening transition words compared to human abstracts.

Among all the analyzed features, lexical diversity appeared as the single most powerful and reliable diagnostic marker, followed closely by sentence length variation. Human authors tend to construct texts with a more diverse rhythm, mixing short and long sentences to manage complex information. Language models, constrained by their autoregressive nature, tend to produce a flatter, more uniform rhythm that is statistically distinct in this dataset. By using these measurable style features, this study built a Random Forest classifier that achieves 95.69% accuracy and a 0.991 AUC-ROC under controlled conditions, balancing precision and recall without the need for black-box neural networks.

This stylometric approach may offer an explainable and transparent complement to existing, opaque AI detection tools [5; 6]. Because the features are human-readable, they could potentially allow for more transparent discussions about authorship rather than relying solely on unexplainable % scores from proprietary software [6]. However, the findings of this study are limited to a specific context. The abstracts were generated by GPT-4o-mini using a single prompt, compared against pre-2022 human abstracts. Whether these same patterns hold across other language models, longer documents, or texts that have undergone human editing remains an open question that future research should address.

As artificial intelligence becomes increasingly integrated into the academic research process, the ability to distinguish between human-written text and machine-generated text remains important to preserving the integrity of the scientific record. Future research could expand on this foundation. Next steps may include testing these stylometric features across specific academic disciplines, applying the models to longer documents like full peer-reviewed manuscripts, and evaluating newer language models to see if these writing patterns remain stable over time. Additional investigation into heavily edited or mixed-authorship texts may be needed to define the detection limits for AI-assisted academic writing [7].

References

[1] Biber, D. (1988). *Variation across Speech and Writing*. Cambridge University Press.. doi.org/10.1017/CBO9780511621024

[2] Hyland, K. (2005). *Metadiscourse on Exploring Interaction in Writing*. London, Continuum.. scholar.google.com/scholar?q=Hyland+Metadiscourse+Exploring+Interaction+Writing+2005

[3] Swales, J. M. (1990). *Genre Analysis, English in Academic and Research Settings*. Cambridge University Press.. doi.org/10.1017/CBO9781139524599

[4] Theocharopoulos, P. C., Anagnostou, P., Tsoukala, A., Georgakopoulos, S. V., Tasoulis, S. K., & Plagianakos, V. P. (2023). Detection of Fake Generated Scientific Abstracts. *2023 IEEE Ninth International Conference on Big Data Computing Service and Applications (BigDataService)*, 33-39.. doi.org/10.1109/BigDataService58306.2023.00011

[5] Guo, B., Zhang, X., Wang, Z., Jiang, M., Nie, J., Ding, Y., Yue, J., & Wu, Y. (2023). How Close is ChatGPT to Human Experts, and a Comparison of Corpus, Evaluation, and Detection. *arXiv preprint 2301.07597*.. arxiv.org/abs/2301.07597

[6] Sadasivan, V. S., Kumar, A., Balasubramanian, S., Wang, W., & Feizi, S. (2023). Can AI-Generated Text be Reliably Detected? *arXiv preprint 2303.11156*.. arxiv.org/abs/2303.11156

[7] Krishna, K., Song, Y., Karpinska, M., Wieting, J., & Iyyer, M. (2024). Paraphrasing Evades Detectors of AI-Generated Text, but Retrieval is an Effective Defense. *Advances in Neural Information Processing Systems*, 36.. arxiv.org/abs/2303.13408

[8] Stribling, J., Krohn, M., & Aguayo, D. (2005). SCIgen - An Automatic CS Paper Generator. *MIT CSAIL*.. pdos.csail.mit.edu/archive/scigen

[9] Mitchell, E., Lee, Y., Khazatsky, A., Manning, C. D., & Finn, C. (2023). DetectGPT for Zero-Shot Machine-Generated Text Detection using Probability Curvature. *Proceedings of the 40th International Conference on Machine Learning (ICML)*.. arxiv.org/abs/2301.11305

[10] Hans, A., Schwarzschild, A., Cheber, V., Nishi, R., Somepalli, G., Goldblum, M., & Goldstein, T. (2024). Spotting LLMs With Binoculars for Zero-Shot Detection of Machine-Generated Text. *Proceedings of the 41st International Conference on Machine Learning (ICML)*.. arxiv.org/abs/2403.19074

[11] McCarthy, P. M. (2005). An Assessment of the Range and Usefulness of Lexical Diversity Measures and the Potential of the Measure of Textual, Lexical Diversity (MTLD). *Doctoral dissertation, University of Memphis*.. proquest.com/docview/305349212

[12] Chafe, W. (1994). *Discourse, Consciousness, and Time on the Flow and Displacement of Conscious Experience in Speaking and Writing*. Chicago, University of Chicago Press.. press.uchicago.edu/ucp/books/book/chicago/D/bo3631492.html

[13] Priem, J., Piwowar, H., & Orr, R. (2022). OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv preprint 2205.01833*.. arxiv.org/abs/2205.01833

[14] Stamatatos, E. (2009). A Survey of Modern Authorship Attribution Methods. *Journal of the American Society for Information Science and Technology*, 60(3), 538-556.. doi.org/10.1002/asi.21001

[15] Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J. D., Dhariwal, P., & Amodei, D. (2020). Language Models are Few-Shot Learners. *Advances in Neural Information Processing Systems*, 33, 1877-1901.. arxiv.org/abs/2005.14165

[16] Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5-32.. doi.org/10.1023/a:1010933404324

[17] Ouyang, L., Wu, J., Jiang, X., Almeida, D., Wainwright, C., Mishkin, P., & Lowe, R. (2022). Training Language Models to Follow Instructions with Human Feedback. *Advances in Neural Information Processing Systems*, 35, 27730-27744.. arxiv.org/abs/2203.02155

[18] McCarthy, P. M., & Jarvis, S. (2010). MTLD, vocd-D, and HD-D: A Validation Study of Sophisticated Approaches to Lexical Diversity Assessment. *Behavior Research Methods*, 42(2), 381-392.. doi.org/10.3758/brm.42.2.381